INDIAN INSTITUTE OF TECHNOLOGY KANPUR

SPA617 — SPASE INTRUMENTATION LABORATORY

EXPERIMENT 1

RF Transmission, Reflections, and Standing Waves

Item	Details
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Course	SPA617 - SPASE INTRUMENTATION LABORATORY
Experiment	RF Transmission, Reflections, and Standing Waves
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1. Aim

To understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

2. Objectives

- Determine the phase velocity and velocity factor of a signal in a coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.
- Study signal reflection from an open-ended cable and from a cable terminated with a 50 Ω load.
- Estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched to a 50 Ω termination, indicating that its characteristic impedance is close to 50 Ω.
- Calculate the inductance and capacitance per unit length using the measured phase velocity and the approximate value $Z_0 \approx 50\ \Omega$.
- Measure the insertion loss and attenuation per unit length of the coaxial cable.

3. Theory

3.1 Transmission lines and reflections

A transmission line guides electromagnetic energy from one point to another. The coaxial cable used in this experiment supports transverse electromagnetic (TEM) propagation. The signal velocity in the cable is lower than the speed of light in vacuum because of the dielectric material between the conductors.

When a travelling wave encounters a load impedance ZL different from the characteristic impedance Z_0 of the cable, part of the wave is reflected toward the source. At the fixed Tee-junction observation point, the incident and reflected waves superimpose. Their relative phase changes as frequency is varied, producing periodic maxima and minima in the measured oscilloscope amplitude.

3.2 Phase velocity measurement

For a cable of physical length ℓ and phase velocity vp, the reflected signal travels a round-trip distance 2ℓ.

$$\tau = 2\ell / vp$$

The round-trip phase is

$$\varphi RT = 4\pi f\ell / vp + \varphi\Gamma$$

For an ideal open circuit, the termination phase shift $\varphi\Gamma$ is zero. Consecutive maxima or consecutive minima are separated by a round-trip phase change of 2π. Hence:

$$\tau = 1 / \Delta f$$

$$vp = 2\ell\Delta f$$

$$VF = vp / c, \text{ where } c \approx 3 \times 10^8 \text{ m/s}$$

3.3 Reflection coefficient

The voltage reflection coefficient is

$$\Gamma = (ZL - Z_0) / (ZL + Z_0)$$

Termination	Reflection coefficient
Open circuit	$\Gamma = +1$
Short circuit	$\Gamma = -1$
Matched load	$\Gamma = 0$

At the Tee junction, the oscilloscope measures the sum of the incident and reflected waves. During a frequency sweep, the maximum and minimum amplitudes are approximately

$V_{max} = |V_{incident}| + |V_{reflected}|$

$V_{min} = |V_{incident}| - |V_{reflected}|$

Therefore, the magnitude of the reflection coefficient can be estimated from

$|\Gamma| = (V_{max} - V_{min}) / (V_{max} + V_{min})$

3.4 Inductance and capacitance per unit length

For a low-loss transmission line:

$v_p = 1 / (L'C')^{1/2}$

$Z_0 = (L'/C')^{1/2}$

Thus, using $Z_0 \approx 50\ \Omega$:

$L' = Z_0 / v_p$

$C' = 1 / (Z_0 v_p)$

3.5 Cable attenuation

Cable attenuation is the reduction in signal level as the wave propagates through the cable. Using the spectrum analyser, the insertion loss is

$IL = P_{ref} - P_{cable}$

and the attenuation per unit length is

$\alpha = IL / \ell$ (dB/m)

4. Components Required

Component	Purpose
RF signal generator	Generate the RF test signal.
Digital storage oscilloscope (DSO)	Measure the signal amplitude at the Tee junction.
Spectrum analyser	Measure reference and cable powers in dBm.
RF Tee connector	Provide the measurement junction.
Coaxial cable under test	Cable whose phase velocity, reflection and attenuation are studied.
Short reference coaxial cable	Provide the reference connection for insertion-loss measurement.
50 Ω termination	Provide an approximately matched termination.
Open-circuit termination	Obtained by leaving the cable end unconnected.
RF adapters and connectors	Complete the RF connections.

5. Instrument Settings and Precautions

- Set the oscilloscope input impedance to 1 MΩ. A 50 Ω input would load the Tee junction and alter the reflection pattern.
- Keep the signal-generator amplitude constant during each frequency sweep.
- Check whether the signal generator displays voltage for a 50 Ω load or a high-impedance load, and use one setting consistently.
- Use a measurable signal level without overdriving the oscilloscope or spectrum analyser.
- Do not connect the signal generator directly to a short circuit without the cable and appropriate source protection.
- Use the same adapters and connector arrangement for reference and cable measurements wherever possible.

6. Experimental Setup

The signal generator, cable under test and oscilloscope are connected through an RF Tee connector. The far end of the coaxial cable is either left open or terminated with a 50 Ω load depending on the measurement.

At a fixed frequency the oscilloscope normally displays a sinusoidal waveform. When the frequency is swept, the measured amplitude varies periodically because the round-trip phase of the reflected signal changes. A matched termination minimizes this amplitude variation.

7. Experimental Procedure

7.1 Part A — Phase velocity and velocity factor

1. Measure and record the physical cable length ℓ.

2. Connect the signal generator, Tee connector, cable and oscilloscope as specified in the laboratory setup.

3. Leave the far end of the cable open.

4. Set the oscilloscope input impedance to 1 MΩ.

5. Apply a sinusoidal signal of constant amplitude.

6. Sweep the frequency slowly over a range containing at least two maxima and two minima.

7. Record the signal amplitude at suitable frequency intervals.

8. Plot measured signal amplitude versus frequency and identify two consecutive maxima or minima.

9. Calculate $\Delta f = |f_2 - f_1|$, using hertz.

10. Calculate $v_p = 2\ell\Delta f$ and $VF = v_p/c$.

7.2 Part B — Reflection coefficient and approximate characteristic impedance

1. Retain the Tee-junction setup and use the same frequency range as Part A.

2. From the open-circuit data, determine $V_{max,open}$ and $V_{min,open}$ over one complete ripple period.

3. Calculate $|\Gamma_{open}| = (V_{max,open} - V_{min,open})/(V_{max,open} + V_{min,open})$.

4. Connect the 50 Ω termination to the far end of the cable.

5. Sweep over the same frequency range without changing the signal-generator amplitude.

6. Record $V_{max,50}$ and $V_{min,50}$.

7. Calculate $|\Gamma_{50}| = (V_{max,50} - V_{min,50})/(V_{max,50} + V_{min,50})$.

8. Compare $|\Gamma_{open}|$ and $|\Gamma_{50}|$.

7.3 Part C — Inductance and capacitance per unit length

Using the measured vp and approximate $Z_0 \approx 50\ \Omega$, calculate $L' = Z_0/vp$ and $C' = 1/(Z_0 vp)$.

7.4 Part D — Cable attenuation

1. Connect the signal generator to the spectrum analyser using the short reference cable.

2. Set a suitable test frequency and input power and keep both fixed.

3. Record the reference power Pref in dBm.

4. Replace the short reference connection with the cable under test without changing the remaining setup.

5. Record the cable power Pcable in dBm.

6. Calculate IL = Pref − Pcable.

7. Calculate $\alpha = IL/\ell$ in dB/m.

8. Repeat at additional frequencies if frequency dependence of attenuation is to be studied.

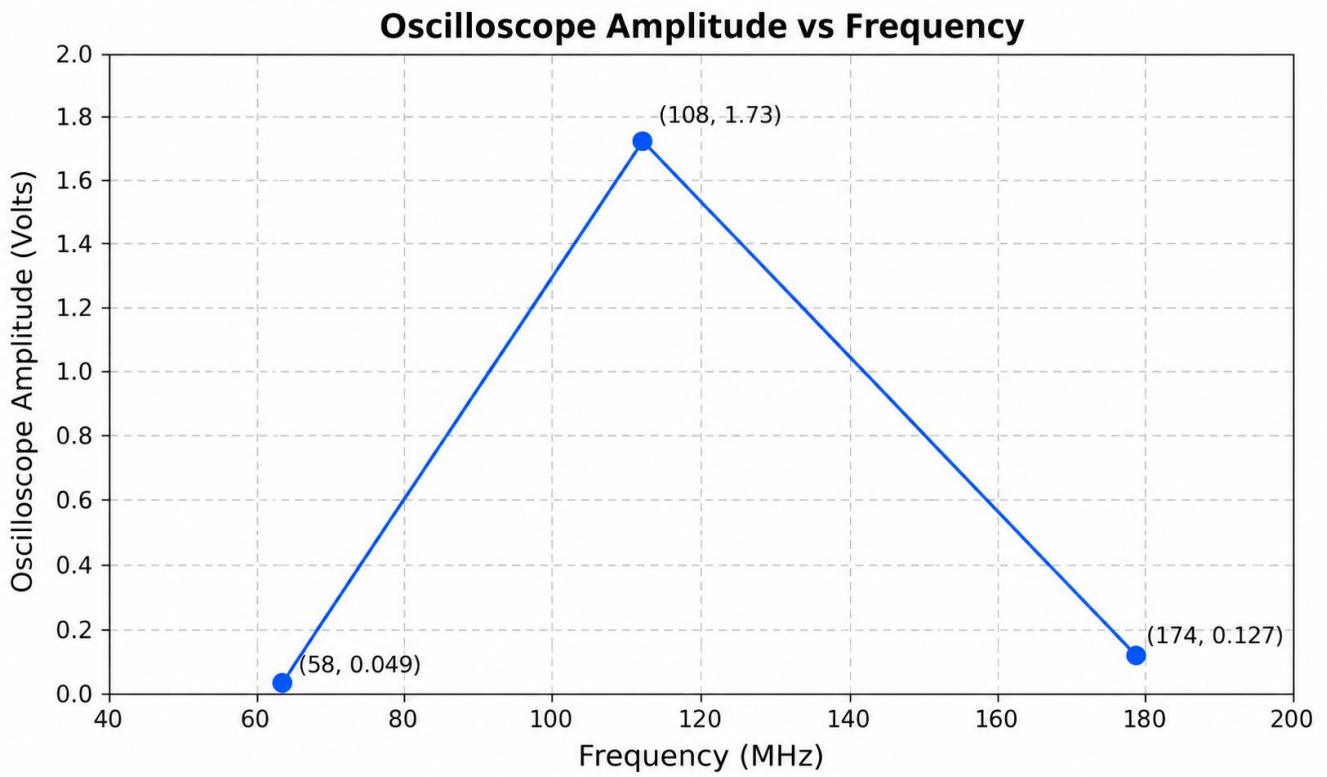
8. Observations & Graphs

Parameter	Sample value
Cable length, ℓ	92.5 cm
Signal-generator amplitude	Constant during sweep
Oscilloscope input impedance	1 MΩ
Open-ended cable	Far end left unconnected
Unknown cable length	306 cm

8.1 Part A — Open-circuit frequency sweep

Frequency (MHz)	Oscilloscope amplitude (Volts)	Comments
58	0.049	First minima
108	1.73	First maxima
174	0.127	Second minima

8.1 Part B – Graph



8.1 Part C — Load-circuit (with 50 Ω) frequency sweep

Frequency (MHz)	Oscilloscope amplitude (Volts)	Comments
10	1.12	First reading
160	0.920	Second reading

8.2 Part D — Reflection measurements

Termination	Vmax (Volts)	Vmin (Volts)	|Γ|
Open circuit	1.73	0.049	0.945
50 Ω load	1.12	0.920	0.098

8.3 Part E — Attenuation measurements

Frequency (MHz)	Pref (dBm)	Pcable (dBm)	IL (dBm)	α (dBm/m)
60	0.00	-0.65	0.65	0.703

*8.4 Part F — Finding Unknown Length (Bonus)

Frequency (MHz)	Oscilloscope amplitude (Volts)	Comments
52	0.120	First minima
87	0.170	First maxima

9. Calculations

9.1 Frequency spacing

For two consecutive maxima, take $f_1 = 84.0$ MHz and $f_2 = 93.9$ MHz.

$\Delta f = |f_2 - f_1| = |174 - 58|$ MHz $= 116$ MHz $= 116 \times 10^6$ Hz

9.2 Phase velocity

$vp = 2\ell\Delta f$

$vp = 2 \times 92.5 \times 10^{-2} \times 116 \times 10^6 \approx 2.146 \times 10^8$ m/s

9.3 Velocity factor

$VF = vp/c = 2.146 \times 10^8 / (3 \times 10^8) \approx 0.7153$

9.4 Reflection coefficient

Open circuit: $|\Gamma open| = (1.73 - 0.049)/(1.73 + 0.049) = 0.945$

50 Ω termination: $|\Gamma 50| = (1.12 - 0.92)/(1.12 + 0.92) = 0.098$

The 50 Ω termination gives a much smaller reflection coefficient than the open circuit, indicating approximate matching of the cable to 50 Ω.

9.4 load Impedance ZL

50 Ω termination: $|\Gamma 50| = (1.12 - 0.92)/(1.12 + 0.92) = 0.098$

$ZL = Z_o (1 - \Gamma 50)/(1 + \Gamma 50) = 50(1 - 0.098)/(1 + 0.098) \approx 41.0747\ \Omega$

9.5 Inductance per unit length

$L' = Z_o/vp$

$L' = 50/(2.146 \times 10^8) \approx 2.33 \times 10^{-7}$ H/m $= 233$ nH/m

9.6 Capacitance per unit length

$C' = 1/(Z_o vp)$

$C' = 1/(50 \times 2.146 \times 10^8) \approx 9.32 \times 10^{-11}$ F/m $= 93.2$ pF/m

9.7 Attenuation

At 60 MHz:

$IL = Pref - Pcable = (0.00) - (-0.65) = 0.65$ dBm

$\alpha = IL/\ell = 0.65/0.925 = 0.703$ dBm/m

*9.8 Finding Unknown Length (Bonus)

We know $v_p \approx 2.146 \times 10^8$ m/s and $VF \approx 0.7153$
And $\Delta f = |f_2 - f_1| = |87 - 52|$ MHz = 35MHz
We know $v_p = 2\ell\Delta f$

Hence, $2\ell = v_p/\Delta f = 2.146 \times 10^8/(35 \times 10^6) \approx 305.7143$ cm

Error = 306 – 305.7143 = 0.2857 cm

10. Results

Quantity	Sample result	Unit
Cable length, ℓ	92.5	cm
Frequency spacing, Δf	116×10^6	Hz
Phase velocity, v_p	2.146×10^8	m/s
Velocity factor	0.7153	
Approximate characteristic impedance, Z_0	≈ 50	Ω
Inductance per unit length, L'	233	nH/m
Capacitance per unit length, C'	93.2	pF/m
Open-circuit $\lvert\Gamma\rvert$	0.945	
50 Ω $\lvert\Gamma\rvert$	0.098	
Attenuation at 60 MHz	0.703	dB/m
Unknown cable length	306	cm

11. Discussion

The open-ended cable produces a periodic variation in the signal amplitude as the frequency is swept. This variation is caused by interference between the incident signal and the signal reflected from the open end. Consecutive maxima or minima correspond to a round-trip phase change of 2π, allowing the cable phase velocity to be obtained from the measured frequency spacing.

For an ideal open circuit the reflection coefficient has magnitude one. In a practical measurement, the observed value may differ from unity because the measured voltage is at the Tee junction and because of source impedance, connector losses, cable losses and measurement uncertainty. With a 50 Ω termination, the amplitude ripple should be substantially reduced if the cable is approximately a 50 Ω transmission line. The comparison of the open and 50 Ω cases therefore provides an experimental check of the approximate characteristic impedance.

The measured phase velocity, together with $Z_0 \approx 50$ Ω, gives the distributed inductance and capacitance of the cable. Spectrum-analyser measurements provide insertion loss, from which attenuation per unit length is obtained.

12. Sources of Measurement Uncertainty

- Uncertainty in the physical measurement of cable length.
- Finite frequency resolution during the frequency sweep.
- Uncertainty in identifying the exact positions/frequencies of maxima and minima (they were fluctuating).
- Oscilloscope amplitude measurement uncertainty and noise.
- Signal-generator amplitude drift during the sweep.

- Connector and adapter mismatch and losses.
- Cable loss and frequency-dependent attenuation.
- Non-ideal behavior of the open and 50 Ω terminations.

13. Conclusion

The experiment demonstrates RF signal transmission, reflection and standing-wave formation in a coaxial transmission line. For the illustrative sample data, the frequency spacing between consecutive minima is 116 MHz for a 92.5 cm cable, giving a phase velocity of approximately 2.146×10^8 m/s and a velocity factor of 0.7153. The 50 Ω termination produces a much smaller reflection coefficient (0.0985) than the open-ended cable (0.7153), supporting the conclusion that the cable is approximately matched to a 50 Ω termination. The corresponding sample distributed parameters are $L' \approx 233$ nH/m and $C' \approx 93.2$ pF/m.

*14. My Understandings

1. If length of the cable is fixed then according to formulae $\varphi RT = 4\pi f\ell / vp + \varphi\Gamma$ (here $\varphi\Gamma$ is zero for open ended), φRT(total phase shift) is only frequency dependent. So, we change the frequencies to get at what points of frequencies, the total phase shift is 0 (for constructive interference) and π (for destructive interference) for getting first minima and maxima.

2. In total impendence of cable, we are not including the resistance of cable in impedance formulae.

3. We are not getting ZL as 50 Ω because of impedance and inductance are dependent on frequency.

4. We have taken first minimums for calculations because at minima same noise will cancel out in destructive interference and we will get more sharp minimums rather then sharp maximums.

5. When we set load as 50 Ω, we are still getting different oscilloscope amplitudes, because of -
   a. The oscilloscope is measuring the signal specifically at the **Tee junction**, not directly along the cable. (extra length at the ends)
   b. The Tee, connectors and adapters can introduce small impedance discontinuities. (they are not perfect)
   c. Cable Attenuation Loss. (Due to dielectric and inductor in that)
   d. Inductor in the cable, with time heated up and increase its resistance therefor Attenuation Loss will increase with time.